

ML-DL-OPs LAB WORK REPORT

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M25CSA001

Colab Link - [M25CSA001_Lab_Work.ipynb](#)

Github Link -

https://github.com/yadityax/MLOps-Aditya_Arun_Kumar_Yadav-M25CSA001.git

Wandb Link -

<https://wandb.ai/m25csa001-indian-institute-of-technology-jodhpur/cifar10-lab2-amp/runs/drlv2y9s?nw=nwuserm25csa001>

Findings and Observations

The observations obtained from training a Convolutional Neural Network (CNN) on the CIFAR-10 dataset for 30 epochs are presented below. The analysis is based on the visualizations logged using **Weights & Biases (W&B)**, including training loss trends, gradient flow, and weight distribution across layers.

1. Training Loss Analysis

From the **training loss vs steps** plot:

- The training loss shows a **monotonically decreasing trend**, starting from a high value (~1.3) and gradually reducing to a very low value (~0.06) by the end of training.
- The sharp decrease in loss during the initial steps indicates that the model **quickly learns low-level features** such as edges and textures.
- As training progresses, the slope of the curve becomes smoother, suggesting **stable convergence**.
- No sudden spikes or oscillations are observed, which implies:
 - The learning rate is well-chosen.
 - Optimization is stable.
 - No exploding gradient issues during training.

Observation:

The consistent decrease in training loss confirms that the CNN is learning effectively and converging without instability.

2. Epoch Progression

The **epoch vs step** visualization shows a linear increase:

- This confirms that logging was correctly synchronized with training iterations.
- The mapping between steps and epochs is uniform, indicating:
 - No skipped batches
 - No early termination
 - Proper DataLoader functioning

Observation:

The training process ran smoothly for all 30 epochs without interruption or data loading issues.

3. Gradient Flow Analysis

The **mean gradient per layer** visualization provides critical insights into backpropagation behavior:

Key Observations:

- Early layers (e.g., `conv1.weight`, `conv1.bias`) exhibit **higher gradient magnitudes**.
- Deeper layers and fully connected layers show **progressively smaller gradients**.
- No layer shows zero gradients, indicating:
 - No dead layers
 - No vanishing gradient problem
- No extremely large gradient spikes are visible, indicating:
 - No exploding gradients

Interpretation:

- Higher gradients in early convolution layers suggest strong learning of foundational visual features.
- Gradual gradient decay across deeper layers is expected and healthy in CNNs.

Observation:

The gradient flow visualization confirms **stable backpropagation**, effective learning across all layers, and correct weight updates throughout training.

4. Weight Distribution Analysis

The **mean weight magnitude per layer** plot shows:

- Convolutional layers have **larger average weights** compared to fully connected layers.
- Fully connected layers (**fc1**, **fc2**) have smaller mean weights, indicating controlled learning.
- Bias parameters generally have slightly higher mean values than corresponding weights, which is typical.

Interpretation:

- Larger weights in convolutional layers indicate strong spatial feature extraction.
- Controlled weight magnitudes across all layers suggest:
 - No weight explosion
 - Proper regularization via optimizer dynamics

Observation:

The weight distribution confirms **healthy parameter scaling** and absence of numerical instability.

5. Relationship Between Gradient Flow and Weight Updates

By jointly analyzing gradient and weight statistics:

- Layers with higher gradients also show more significant weight updates.
- Fully connected layers exhibit smaller updates, which is expected as they operate on higher-level representations.
- No layer appears frozen or over-active.

Observation:

The alignment between gradient magnitudes and weight updates confirms that **all layers are learning effectively**, and no part of the network is under-trained or over-trained.

6. FLOPs

The computational complexity of the model was evaluated using the **THOP library**. With a standard input size of **$1 \times 3 \times 32 \times 32$** , the model requires **6.85 million MACs (MMACs)**, corresponding to approximately **13.7 million FLOPs**, and contains **1.07 million trainable parameters**. The majority of the computational cost arises from the convolutional layers, while fully connected layers contribute comparatively less.

7. Overall Model Behavior

Based on all visualizations:

- The model converges smoothly.
- Gradient propagation is stable.
- Weight updates are balanced across layers.
- The optimization process is well-behaved.

Final Conclusion:

The CNN model trained on CIFAR-10 demonstrates **stable learning, effective gradient propagation, and well-regulated weight updates**. The visualizations validate that the chosen architecture, optimizer, learning rate, and batch size are appropriate for the task.